Customer Segmentation Analysis Using Rule-Based Classification

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Project Objective

The objective of this project is to analyze customer data and segment customers based on their income and spending behavior.

Customer segmentation helps businesses understand different customer groups and develop targeted marketing strategies to improve sales and customer engagement.

Dataset Description

The dataset contains customer information with the following details:

- Total Records: 200
- Features Used:
  - Age
  - Annual Income (₹000)
  - Spending Score (1–100)

The Spending Score represents customer purchasing behavior and engagement level.

Methodology

This project uses a **Rule-Based Segmentation Approach**.

Customers were segmented using rule-based conditions derived from annual income and spending score.

Segmentation Logic:

Segment Name	Criteria
Premium	High income & High spending
Budget	Low income & Low spending
Careless	High income & Low spending
Value Seeker	Low income & High spending

This classification helps identify different purchasing behaviors across customer groups.

Tools & Technologies Used

- Python
- Jupyter Notebook
- Pandas (Data manipulation)
- Matplotlib & Seaborn (Visualization)

Data Processing Steps

1. Imported dataset using Pandas
2. Checked for missing values
3. Applied conditional rules for segmentation
4. Created new column: “Customer Segment”
5. Generated visualizations to analyze distribution

Visualization & Analysis

A scatter plot was created to visualize customer segments based on:

- Annual Income
- Spending Score

Different colors were used to represent each segment.

This visualization clearly shows how customers are grouped according to their purchasing behavior.

Files Included in the Project Folder:

- customer_data.csv: Raw dataset
- segmented_customers.csv: Final data with segment labels
- segmentation_analysis.ipynb: Python notebook
- README.txt: Quick project summary
- Customer_Segmentation_Report.pdf: This report

Key Insights:

• Premium customers represent high-income individuals with strong purchasing behavior.
• Budget customers show low spending and are likely price-sensitive.
• Careless customers have high income but low spending engagement.
• Value Seekers show strong spending behavior despite lower income.

Conclusion:

This project demonstrates how customer segmentation can help businesses identify meaningful patterns in purchasing behavior.

By using rule-based classification, businesses can design targeted marketing strategies and improve customer retention.